

E. Effect on Efficiency, Competition, and Capital Formation

1. Efficiency

The amendments will improve price efficiency, namely the degree to which the price of a stock reflects its fundamental value. The improvement in price efficiency is expected largely to come through the reduction in the tick size and the reduction of the access fee cap. The acceleration of portions of the MDI Rules could also increase price efficiency, but those effects are largely to accelerate the economic impact already anticipated in the MDI Rules.

Lowering the tick size for some NMS stocks with prices equal to or greater than \$1.00, as well as lowering the access fee cap for all stocks to 10 mils for stocks with prices equal to or greater than \$1.00, or to 0.10% for stocks with prices lower than \$1.00, will increase price efficiency.¹⁷²⁹ The reduction in the tick size for some stocks along with the reduction of the access fee cap for all stocks will lower transaction costs.¹⁷³⁰ When trading becomes less costly, market participants have an increased incentive to gather more information because doing so is more profitable.¹⁷³¹ Gathering more information and trading on that information means that prices are more reflective of the fundamental value of the firm. Consequently, for stocks that receive an improvement in market quality due to the lower tick size or the reduction in the access fee, the Commission expects an improvement in price efficiency.¹⁷³²

¹⁷²⁹ See NASAA Letter at 9, and Vanguard Letter at 6.

¹⁷³⁰ As discussed in section VII.B, the reduction in the fee cap will primarily lower trading costs and distortions for stocks that are tick-constrained; for stocks that are not tick-constrained, the reduction in the fee cap may lead to wider quoted spreads which will offset the lower fee. See supra note 1508 for a discussion of the fee cap's contribution to liquidity distortions for tick-constrained stocks; see also supra section VII.D.2.c on the effect of the fee cap reduction for stocks that are not tick-constrained.

¹⁷³¹ See, e.g., Dixon, supra note 1277 for a discussion of this concept in the context of short selling.

¹⁷³² Id.

The Commission further expects that quoted spreads will better reflect the cost of liquidity as a result of these amendments. As discussed in section VII.B.3,¹⁷³³ high access fees and rebates can distort liquidity supply and demand.¹⁷³⁴ The tick acts as a price floor that prevents the spread from reflecting the true cost of liquidity; likewise, access fees—and the rebates they fund—further distort quoted spreads by taxing liquidity demand and subsidizing liquidity supply at this price floor. By reducing both the tick and the access fee cap, quoted spreads will better reflect the cost of liquidity and allow for more efficient liquidity provision.

Some commenters expressed the concern of diminished intra-tick pricing from reduced access fees.¹⁷³⁵ As shown in the Proposing Release, the NYSE, Nasdaq, and Cboe exchange families each operate both a maker-taker venue and an inverted venue.¹⁷³⁶ Variation in fee and rebate schedules across trading venues effectively allow for intra-tick pricing.¹⁷³⁷ However, that intra-tick pricing via access fees and rebates is an imperfect solution to the problem of a stock having a tick that is too large.¹⁷³⁸ Reducing the access fee cap can reduce the degree to which maker-taker trading centers can offer effective intra-tick pricing which could potentially lead to

¹⁷³³ See also Proposing Release, note 11, at 80328: “If tick sizes were infinitely small, and absent other distortions, then fees and rebates would not affect the cost of trading because markets would simply adjust quotes by the amount of the rebate such that the spread with rebates included is the same. However, current U.S. equity markets differ from this frictionless construct because there is a finite tick. In this environment, and particularly for stocks with narrower spreads, high access fees and rebates can distort liquidity supply and demand by artificially increasing the cost of taking liquidity and the revenue to providing liquidity. This dynamic creates an environment with too much liquidity supply relative to liquidity demand.”

¹⁷³⁴ One commenter stated that the Commission did not explain these distortions, see Virtu Letter II at 16. The subsequent discussion (as well as the discussion surrounding note 1508, supra) explains the distortions.

¹⁷³⁵ See also Citadel Letter I at 16, CCMR Letter at 26.

¹⁷³⁶ See 87 FR 80313.

¹⁷³⁷ See also supra note 1128.

¹⁷³⁸ See Quarter Penny Tick, supra note 1127.

pricing distortions – i.e., less efficient prices.¹⁷³⁹ However, the ability for markets to establish efficient prices will increase overall due to the reduced tick size for stocks with TWAQS less than or equal to \$0.015. The reduced tick size reduces the need for intra-tick pricing by providing a finer pricing grid on which market participants can submit orders. Specifically, a tick size of \$0.005 with an access fee and rebate of 10 mils provides an effective pricing grid with price pints at every \$0.005 and plus or minus \$0.001, which is finer than the current pricing grid with a tick size of \$0.01 and an access fee/rebate of \$0.003.¹⁷⁴⁰ For stocks not subject to the reduced tick size, the Commission acknowledges some reduction in the ability to price intra-tick. However, the efficiency loss is limited due to the fact that, by definition, these stocks have sufficiently wide average spreads, and thus sufficient ticks within the spread, to avoid qualifying for the lower tick size. For these stocks, there is less of a need to price intra-tick.

The reduction in the tick is expected to reduce the cost of transacting on exchanges, which will result in an increase in orders being sent to lit markets.¹⁷⁴¹ The reduction in the access fee cap is expected to widen quoted spreads, and the quote differential between lit exchanges and

¹⁷³⁹ See supra note 1128 for an explanation of how variation in fee and rebate schedules across trading venues can increase price fidelity by allowing for more effective intra-tick pricing. Reducing the access fee cap reduces this effective intra-tick pricing by limiting the degree to which fee and rebate schedules can differ from one another.

¹⁷⁴⁰ Under the preexisting Rule 610 fees, maker-taker venues can offer liquidity demanders net prices that are 30 mils worse than quoted prices; inverted venues, in contrast, tend to offer net prices that are 30 mils better than quoted prices, creating a grid of price points that are 40 to 60 mils separated from each other. E.g., the net prices available at a maker-taker venue may be \$10.003, \$10.013, etc., while the net prices available at an inverted venue may be \$9.997, \$10.007, etc.; the price points are 40 to 60 mils apart from each other. Under the reduced fee cap, maker-taker venues will only be able to offer liquidity demanders net prices that are 10 mils worse than quoted prices; to maintain a grid of price points that are 50 mils apart, an inverted venue could offer a net price that is 40 mils better than the quoted price by instituting a 40 mil rebate for liquidity takers. E.g., the net prices available at a maker-taker venue will be \$10.001, \$10.013, etc.; with a 40 mil rebate, inverted venues could offer net price points of \$9.996, \$10.006, etc., so that the price points are 50 mils apart from each other. See also Quarter Penny Tick, supra note 1127, for a discussion of this concept.

¹⁷⁴¹ See infra section VII.E.2.a and VII.E.2.b for additional discussion. Specifically see discussion surrounding infra note 1752.

other trading venues may widen. This need not, however, lead to lower demand for lit liquidity, as the access fee, which currently disincentivizes investors to access liquidity on exchanges, will also be lower.¹⁷⁴² Thus, the net effect of the rules is an increase in orders on lit markets and an improvement in price efficiency.

Some commenters stated that a finer tick size could cause information leakage. Large orders may need to be divided into smaller orders due to the fragmentation of liquidity across multiple price levels; executing a large order may therefore reveal more information, which could increase price impact and trading costs.¹⁷⁴³ The Commission acknowledges that information leakage may, in turn, reduce incentives to collect information ex-ante and thereby reduce price efficiency. However, on balance, the effect of the amendments to increase price efficiency by, on average, reducing trading costs. This is because the cost of information leakage primarily shows up in the form of wider spreads when a large trade is anticipated and higher costs of trading for large orders. If information leakage was a primary—rather than mitigating—effect of the smaller tick size, then in the TSP analysis discussed in section VIII.D.1.b.ii spreads would not have narrowed and round-trip trading costs would not have fallen for stocks with narrow spreads when the tick size was reduced. But this is not what that analysis found; thus, while increased information leakage could be a factor mitigating the reduction in spreads due to the smaller tick size, the net effect is expected to be lower trading costs for stocks receiving the

¹⁷⁴² See, e.g., IEX Letter VI at 1-6.

¹⁷⁴³ See supra notes 1219, 1220 and 1398, as well as related discussion in sections VII.D.1.b.i and VII.D.1.e. The Commission acknowledges that tick-constrained stocks receiving a tick reduction may experience an increase in execution costs for sufficiently large orders—evidence from the TSP suggests that this occurs for orders that are quite large (approximately 50 round lots). See discussion surrounding note 1284.

smaller tick size. Additionally, the commenter's concerns are mitigated by the fact that the amendments do not include the proposed smaller tick sizes (0.001 and 0.002).

Lowering access fees also increases the efficiency with which the quote conveys information regarding the cost of liquidity. As discussed in section VII.D.2.d, access fees that fund rebates contribute to complexity and lack of transparency in markets because they separate both the true cost of demanding liquidity and the proceeds from supplying liquidity, as represented by the quoted half-spread. Reducing the wedge between the spread and the true cost also reduces conflicts of interest between broker-dealers and their customers. Multiple commenters stated that a lower access fee cap would help mitigate the conflict of interest because lowering the access fee is expected to reduce rebates available and thus the incentive to route based on rebates instead of execution quality.¹⁷⁴⁴

Making fees and rebates determinable at the time of execution, along with the reduction of the access fee cap could also increase price efficiency by helping minimize potential conflicts of interest.¹⁷⁴⁵ Fees and rebates create a potential conflict for a broker in situations where incentives related to transaction fees, which are paid by the broker, potentially conflict with incentives to obtain execution quality, which may affect the customer.¹⁷⁴⁶ This conflict, if acted on, can lead to inefficient order routing and worse transaction outcomes for customers;¹⁷⁴⁷ it can

¹⁷⁴⁴ See RBC Letter at 4 stating ("we believe that lowering the access fee cap would lead to a reduction in broker conflicts of interest..."); NASAA Letter at 9 (stating "reducing access fee caps could help reduce incentives for broker-dealers to route orders to trading venues that benefit those broker-dealers (such as venues in which a broker-dealer is rebated), but may provide suboptimal execution to the detriment of the broker-dealer's customers."). See also Themis Letter at 7; supra section VII.D.3.

¹⁷⁴⁵ One commenter stated that they agreed with the Commission's analysis of these effects, see Council of Institutional Investors Letter at 5; see also Themis Letter at 7.

¹⁷⁴⁶ See Vanguard Letter at 6. See also supra section VII.D.2 for additional discussion of access fees.

¹⁷⁴⁷ See Retirement Coalition Letter at 2.

also lead to an inefficient incorporation of information into stock prices, harming market efficiency. Making access fees determinable at the time of execution will enhance efficiency by providing market participants with certainty concerning the fees that they will be charged per transaction. This certainty could also allow broker-dealers to examine their own best-execution performance more efficiently. Greater certainty about fees and rebates in advance of routing an order could also increase the efficiency of the broker-dealers' best execution assessments by providing them with greater certainty about the full cost of a transaction when executing the order. Additionally, to the extent that determinable fees make it easier for broker-dealers to communicate fees and transmit them to end customers, doing so could help eliminate distortions that might occur due to potential conflicts of interest. However, to the extent that exchanges are not able to as effectively replicate some incentives that were based on using historical activity,¹⁷⁴⁸ requiring these fees to be determinable at time of execution may reduce efficiency.

Accelerating the addition of odd-lot information to NMS data and the inclusion of information relating to the best odd-lot quote will realize some of the price efficiency benefits articulated in the MDI Rules at an earlier date, providing improved price efficiency earlier than anticipated in the MDI Rules. Specifically, research suggests that adding information on the shares available at price levels inside the NBBO may improve price efficiency.¹⁷⁴⁹ Currently only market participants who subscribe to proprietary data feeds can view the odd-lot information and thus adjust trading strategies and decisions based on that information.

¹⁷⁴⁸ See supra section VII.D.3 for further discussion on basing fees and rebates on historical activity.

¹⁷⁴⁹ See Robert P. Bartlett et al., The Market Inside the Market: Odd-Lot Quotes, REV. FIN. STUD. (Sep. 19, 2023), available at <https://doi.org/10.1093/rfs/hhad074>.

Expanding the SIP feeds to include odd-lot information will provide new information to those investors who subscribe to the SIP data but do not subscribe to proprietary data feeds.¹⁷⁵⁰

2. Competition

a. Modification of Rule 612

The amendments will promote competition both on price on a given venue and across venues. This will occur because the amendments will allow liquidity providers to compete at more price points on exchange. By limiting the affected stocks to those with low spreads, the amendments ameliorate possible effects of pennyning which may accompany a finer pricing grid.¹⁷⁵¹

In addition, the amendments will improve the efficiency of on-exchange trading, allowing exchanges to better compete with off-exchange market makers. Empirical evidence suggests that, on average, relaxing tick constraints leads to volume moving onto exchanges primarily by improving market quality on the exchanges.¹⁷⁵² Research suggests that this occurs both because of the reduction in transaction costs resulting from a finer pricing grid, and also

¹⁷⁵⁰ There is the possibility that competing consolidators may not choose to distribute odd-lot information (because the MDI Rules do not require them to do so), in which case the positive effects of including odd-lots in NMS data on price efficiency will be lost. This outcome is unlikely because the odd-lot information is valuable in terms of having information relevant to stock prices, see Bartlett et al., id. and the alternative to odd-lot information from the competing consolidators would be to subscribe to all of the proprietary data feeds, which is expensive. Given that there will be significant demand for the odd-lot information, competing consolidators will therefore offer the data.

¹⁷⁵¹ See supra section VII.D.1.b for additional discussion of the effects of pennyning.

¹⁷⁵² See two industry studies attached to MEMX Letter at 43-63 and 64-70. These studies examine the relaxation of tick constraints following reverse splits and find that the reduction in on-exchange trading costs results in an increase in on-exchange trading volume. See also Panel A of Table 2 in Bidisha Chakrabarty, et al., Tick Size Pilot Program and Price Discovery in US Stock Markets, 59 J. FIN. MKT. 100658 (2022). This academic study uses the TSP and finds that an increase in tick constraints results in an increase in off-exchange trading volume. See also Amy Kwan et al., Trading Rules, Competition for Order Flow and Market Fragmentation, 115 J. FIN. ECON. 330 (2015). This academic study examines the change in the tick from \$0.01 to \$0.0001 for orders priced in the vicinity of \$1.00, and finds that more volume is executed on exchanges when the trade price dips below \$1.00 and is therefore subject to the smaller tick.

because a tick that is too wide creates long queues for limit order execution and increase the incentives to send orders off-exchange.¹⁷⁵³

The increase in message traffic expected from the reduction in the tick sizes for certain stocks will result in a mild increase in costs to process such traffic for those who receive the relevant data feeds.¹⁷⁵⁴ Because such technological costs are largely fixed, and do not depend on the size of the broker-dealer, this could disadvantage smaller broker-dealers in the market to provide broker-dealer services to investors.

Some commenters stated that variable tick sizes could increase confusion among investors trading on-exchange, thereby driving orders off-exchange.¹⁷⁵⁵ The potential for investor confusion is addressed generally in section VII.D.1.d. To the point about confusion due to a smaller tick size driving order flow off of exchanges, this is unlikely. The ability to trade at finer price points, and the reduced need to wait in the queue should contribute to on-exchange trading, not off-exchange trading. Indeed, relaxing of tick constraints has been associated empirically with volume moving on-, not off-exchange.¹⁷⁵⁶

One commenter asked the Commission to consider the competitive effects of Rule 612 on stocks that have similar quoted spreads but fall just on either side of the threshold, specifically similar ETPs that may have quoted spreads that are similar but fall on either side of the threshold and so receive different tick sizes.¹⁷⁵⁷ The commenter considers two issuers, Issuer A and Issuer B, and explains that a narrower tick sizes for Issuer A could attract more liquidity to Issuer A's

¹⁷⁵³ Id.

¹⁷⁵⁴ See supra section VII.D.1.c.

¹⁷⁵⁵ See Themis Letter at 6 and Cboe Letter II at 7.

¹⁷⁵⁶ See supra note 1752.

¹⁷⁵⁷ See SIFMA Letter II at 41.

stock and less liquidity to Issuer B's stock. Once Issuer A's stock attracts more liquidity, its spreads could potentially narrow further, perpetuating a cycle in which Issuer B's shares are unable to catchup to Issuer A. The Commission, however, does not expect significant competitive effects in this situation. While the evidence suggests that stocks with quoted spreads less than the threshold will, on average, benefit from the lower tick, those benefits attenuate as spreads widen.¹⁷⁵⁸ Thus, for stocks or ETPs with spreads right at the threshold, the differential effect of the smaller tick size may be relatively small.

b. Lower Access fee Caps

The amendments to Rule 610(c) reducing the access fee cap will have varying effects on competition between trading venues as well as competition between broker-dealers. The Commission does acknowledge that it would limit the ability of exchanges to differentiate themselves from other exchanges on the basis of their pricing schedules; however, the Commission expects that exchanges will continue to set fees and rebates at or near the access fee cap. A lower access fee cap mechanically reduces the range over which pricing tiers can vary, potentially reducing the economic differences between pricing tiers thereby reducing the benefits from routing order flow for the purpose of qualifying for one tier over another. This can reduce the competitive wedge between high and lower volume broker-dealers due to volume discounts making it easier for lower volume broker-dealers to compete with larger volume broker-dealers.

Commenters disagreed regarding the effects of the 10 mils cap relative to the 15 mils/30 mils alternative in terms of the competitive dynamics between on- and off-exchange venues, with some commenters arguing that reducing the access fee cap would cause a shift to on-exchange

¹⁷⁵⁸ See supra section VII.D.1.

trading while others arguing it would cause a shift to off-exchange trading.¹⁷⁵⁹ The Commission's discussion in section VII.D.2.c suggests that it is unlikely that significant activity will be driven off-exchange, and it is possible that activity may come on-exchange as a result of the lower access fee cap. Moreover, the lower access fee cap will improve competition relative to the 15 mils/30 mils alternative in that it will reduce information asymmetries among investors, better aligning displayed prices with the actual costs of transactions.¹⁷⁶⁰

Some commenters expressed that the reduction in access fees will impede exchange competition by reducing their ability to offer differentiated pricing.¹⁷⁶¹ One commenter stated that this will particularly disadvantage new exchanges with limited opportunities for differentiation.¹⁷⁶² Another commenter stated that the inability to differentiate based on fees and rebates will lead volume to congregate on the listing exchange to the detriment of non-listing exchanges.¹⁷⁶³

The Commission acknowledges that lowering the access fee cap will mechanically limit the extent to which exchanges can potentially differentiate themselves based on varying pricing schedules and diminish their ability to compete on the basis of their pricing schedules. However it is not clear if the amount of differentiation or degree of competition will diminish because exchanges do not appear, in the markets today, to be competing on the basis of offering

¹⁷⁵⁹ We focus on the competitive effect of the access fee cap relative to the alternative, as failure to reduce the access fee cap in the presence of the tick constraint would lead to a loss of price coherence. For commenters stating that reducing the access fee cap would increase off-exchange volume, see, e.g., Nasdaq Letter II at 4 and Cboe Letter IV at 2-3. For commenters stating that reducing the access fee cap would increase on-exchange volume, see IEX Letter VI at 7.

¹⁷⁶⁰ See IEX Letter VI at 4-6.

¹⁷⁶¹ See Fidelity Letter at 14, Virtu Letter II at 10, Cboe Letter II at 8, and Citadel Letter I at 24.

¹⁷⁶² See Virtu Letter II at 19.

¹⁷⁶³ See Cboe Letter II at 7.